Solutions

1. Mass percentage (W/W) = $\frac{\text{Mass of the compount in solution}}{\text{Total mass of the solution}} \times 100$

eg 10% glucose in water → 10g of glucose is dissolved in 90g of wat
→ 100 g of solution

2. Volume percentage (V/V) = $\frac{\text{Volume of the component}}{\text{Total volume of solution}} \times 10$

eg 35% (V/V) Ethanol → 35 ml of ethanol in 65 ml of water
→ 100 ml of solution

3. Mass by volume percentage (W/V)

$= \frac{\text{Mass of solute}}{\text{Total volume of solution}} \times 100$

eg 10% NaCl (W/V) → 10g of NaCl in 100 ml of solution

4. Parts per million (ppm) = $\frac{\text{Number of parts of the component}}{\text{Total number of parts of all component of solution}} \times 10^6$

eg Pollutant
Dissolved O_2

5 Mole fraction (X) = $\frac{\text{Number of moles of the component}}{\text{Total number of moles of all component}}$

n_A, n_B $\boxed{X_A = \frac{n_A}{n_A + n_B}}$ $\boxed{X_B = \frac{n_B}{n_A + n_B}}$ $X_A + X_B = 1$

FORMULA SHEET

Sourabh Raina

6. Molarity (M) = $\frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$

→ Molarity varies with temperature

7 Molality (m) = $\frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$

→ Molality is independent of temperature

8. Henry's law

$p \propto X$ $\boxed{p = k_H X}$ p = partial pressure of gas

Henry's constant X = Mole fraction of gas in solution

9. Raoult's law

Volatile liquids $P_1 \propto X_1$ $P_2 \propto X_2$

$P_1 = P_1^\circ X_1$ $P_2 = P_2^\circ X_2$

P_1° and P_2° is vapour pressure of pure component 1 and 2.

$\boxed{P_{Total} = P_1 + P_2}$

$P_1 = y_1 P_{Total}$

$P_2 = y_2 P_{Total}$

y_1 and y_2 mole fraction of 1 and 2 in vapour phase.

Non volatile solute $\boxed{P_S = P^\circ X_1}$ → Mole fraction of solvent

→ Vapour pressure of pure solvent

10. Ideal solution

→ A-A + B-B = A-B $\Delta V_{mix} = 0$

→ $P_T = P_A + P_B$ $\Delta H_{mix} = 0$

11 Non ideal solution

Positive deviation	Negative deviation
$A-A, B-B > A-B$	$A-A, B-B < A-B$
$P_T > P_A + P_B$	$P_T < P_A + P_B$
$\Delta H_{mix} = +ve$	$\Delta H_{mix} = -ve$
$\Delta V_{mix} = +ve$	$\Delta V_{mix} = -ve$

Colligative properties

12. Relative lowering of Vapour Pressure

$$\boxed{\frac{P_1^\circ - P_1}{P_1^\circ} = X_2}$$

$P_1^\circ \rightarrow$ V.P of Pure solvent

$P_1 \rightarrow$ V.P of solution

$X_2 \rightarrow$ Mole fraction of solute

$$\frac{P_1^\circ - P_1}{P_1^\circ} = \frac{n_2}{n_1+n_2}$$

For dilute solution $n_1 >> n_2$

$$\boxed{\frac{P_1^\circ - P_1}{P_1^\circ} = \frac{n_2}{n_1}}$$

13 Elevation in Boiling point

$$\Delta T_b = K_b m$$

$$\boxed{\Delta T_b = \frac{K_b \times W_2 \times 1000}{M_2 \times W_1}}$$

$\Delta T_b \rightarrow$ elevation in Boiling point

$K_b \rightarrow$ Molal elevation constant unit $K\,kg\,mol^{-1}$

$W_2 \rightarrow$ Mass of solute

$W_1 \rightarrow$ Mass of solvent

$M_2 \rightarrow$ Molar mass of solute

14. Depression in Freezing point

$$\Delta T_f = K_f \times m$$

$$\boxed{\Delta T_f = \frac{K_f \times W_2 \times 1000}{W_1 \times M_2}}$$

$K_f \rightarrow$ Molal depression constant $K\,kg\,mol^{-1}$

$W_2 \rightarrow$ Mass of solute (g)

$W_1 \rightarrow$ Mass of solvent (g)

$M_2 \rightarrow$ Molecular mass of solute

15. Osmotic Pressure (π)

$$\pi = CRT$$

$$\pi = \frac{n_2}{V} RT$$

$$\boxed{\pi = \frac{W_2 \times R \times T}{M_2 \times V}}$$

$W_2 \rightarrow$ Mass of solute

$M_2 \rightarrow$ Molar mass of solute

$V \rightarrow$ Volume of solution (l)

16 Van't Hoff factor (i)

$$i = \frac{\text{Normal molar mass}}{\text{observed molar mass}}$$

$$i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$$

NaCl , K_2SO_4

$i = 2$ $i = 3$

Ethanoic acid in Benzene

$i = 0.5$

$$\boxed{\frac{\Delta P}{P} = i X_2}$$

$$\boxed{\Delta T_f = i K_f m}$$

$$\boxed{\pi = iCRT}$$

$$\boxed{\Delta T_b = i K_b m}$$

$i > 1$ Dissociation

$i < 1$ Association

$i = 1$ No dissociation or association

$\alpha \rightarrow$ degree of dissociation

$n \rightarrow$ no of particles dissociates

Dissociation $\rightarrow$ $\boxed{i = 1 + (n-1)\alpha}$

$\beta \rightarrow$ degree of association

$n \rightarrow$ no of particle associates

Association $\boxed{i = 1 - \beta + \frac{\beta}{n}}$

Electrochemistry

Daniell cell → $Zn(s) | Zn^{2+}_{(aq)} || Cu^{2+}_{(aq)} | Cu$ $E_{cell} = 1.1V$

$E_{ext} < 1.1V$	$E_{ext} = 1.1V$	$E_{ext} > 1.1V$
$\bar{e}$ flow Zn to Cu	No flow of electron	$\bar{e}$ flow Cu to Zn electrolytic cell

Cell potential

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{Anode}$$

E° = standard electrode potential

Nernst equation

$$aA + bB \xrightarrow{n\bar{e}} cC + dD$$

$R = 8.314\ JK^{-1} mol^{-1}$

$F = 96500\ C mol^{-1}$

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \log Q_c$$

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \log \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

298k

$$\boxed{E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[C]^c [D]^d}{[A]^a [B]^b}}$$

$Zn(s) + Cd^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cd(s)$ $E_{cell} = E^\circ_{cell} - \frac{0.059}{2} \log \frac{[Zn^{2+}]}{[Cd^{2+}]}$

At equilibrium

$$E^\circ_{cell} = \frac{2.303\,RT}{nF} \log k_c$$

At 298k

$$E^\circ_{cell} = \frac{0.0591}{n} \log k_c$$

Gibbs free energy

$$\Delta G^\circ = -nFE^\circ_{cell}$$

$$\Delta G^\circ = -2.303\,RT \log k_c$$

Conductance of electrolytic solution

Resistance $R = \rho \frac{l}{A}$ unit → Ω

$\rho = R \times \frac{A}{l}$ unit → Ω cm

Resistivity

Conductance $G = \frac{1}{R} = \frac{1}{\rho} \times \frac{A}{l}$

$G = k \times \frac{A}{l}$ Unit → Ω^{-1} siemens (S)

Conductivity (specific conductance)

$$\boxed{k = G \times \frac{l}{A}}$$

unit → $S\,cm^{-1}$

limiting molar conductivity ($\Lambda^\circ m$)

→ Molar conductivity at infinite dilution

Strong electrolyte → Debye Huckel equation

$$\Lambda m = \Lambda^\circ m - A\sqrt{c}$$

A → Constant depend on type of electrolyte

Kohlrausch law

$$\Lambda^\circ m = V_+ \lambda^\circ_+ + V_- \lambda^\circ_-$$

V_+ → No of cation

V_- → No of anion

$$\Lambda^\circ MgCl_2 = \lambda^\circ Mg^{2+} + 2\lambda^\circ Cl^-$$

Dissociation constant

$$Ka = \frac{C\alpha^2}{1-\alpha}$$

Degree of dissociation

$$\alpha = \frac{\Lambda m}{\Lambda^\circ m}$$

Electrolytic cell

Faraday's first law of electrolysis

$W \propto Q$

$W = ZIT$

$$W = \frac{M \times I \times T}{nF}$$

M → Molar mass

n → electron

I → Current (A)

t → Second

$F = 96500C$

Faraday's 2nd law of electrolysis

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

E = Equivalent Weight

$Na^+ = \frac{23}{1} = 23$

$Mg^{2+} = \frac{24}{2} = 12$

$Al^{3+} = \frac{27}{3} = 9$

Chemical Kinetics

$$R \longrightarrow P$$

$[\ ] \rightarrow mol\,l^{-1}$

Rate of disappearance of R $= -\dfrac{\Delta[R]}{\Delta t}$

Rate of appearance of P $= \dfrac{\Delta[R]}{\Delta t}$

Unit $\rightarrow mol\,l^{-1}s^{-1}$

→ stiochiometric coefficient same

$PCl_5 \rightarrow PCl_3 + Cl_2$ Rate of reaction $= -\dfrac{\Delta[PCl_5]}{\Delta t} = +\dfrac{\Delta[PCl_3]}{\Delta t}$

$= \dfrac{\Delta[Cl_2]}{\Delta t}$

→ stiochiometric coefficient different

$2N_2O_5 \longrightarrow 4NO_2 + O_2$ Rate of reaction $= -\dfrac{1}{2}\dfrac{\Delta[N_2O_5]}{\Delta t} = \dfrac{1}{4}\dfrac{\Delta[NO_2]}{\Delta t}$

$= \dfrac{\Delta[O_2]}{\Delta t}$

Rate expression and Rate constant

$aA + bB \rightarrow cC + dD$

$k \rightarrow$ Rate constant.

$$\boxed{Rate = k[A]^x[B]^y}$$

$x, y \rightarrow$ determined experimentally

Order of reaction

$Rate = k[A]^x[B]^y$

$x + y =$ order

zero, fraction, decimal

Unit of Rate constant

zero order reaction $\rightarrow mol\,l^{-1}s^{-1}$

First order reaction $\rightarrow s^{-1}$

Second order reaction $\rightarrow mol^{-1}\,l\,s^{-1}$

Molecularity

No of reacting species in elementary reaction

1, 2, 3 → Trimolecular

1 → Unimolecular

2 → Bimolecular

Zero order reaction

$$k = \frac{[R]_0 - [R]}{t}$$

$$t_{1/2} = \frac{[R]_0}{2k}$$

First order reaction

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

$$t_{1/2} = \frac{0.693}{k}$$

Arrhenius equation

$$k = A e^{-Ea/RT}$$

A → Arrhenius factor

Ea → Activation energy

R → Gas constant

$$\log k = \log A - \frac{Ea}{2.303RT}$$

→ k_1 and k_2 are rate constant at temperature T_1 and T_2

$$\log \frac{k_2}{k_1} = \frac{Ea}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Collision theory

$$\text{Rate} = P Z_{AB} e^{-Ea/RT}$$

Z_{AB} → Collision frequency

P → Steric factor or probability factor